ELECTRIC VEHICLE MARKET PERFORMANCE & INSIGHTS — FINAL DOCUMENTATION

- **Author:** Mufaddal Bag Wala

- **Dataset Analyzed:** ev_market_data.csv (30 Global EV Models | 3,000,000 Annual Sales)
- **Tableau Public Link:** [https://public.tableau.com/app/profile/mufaddal.bagwala/viz/EVMarketCapstoneProject/Dashboard1]
- **Demo Video Link:** [https://drive.google.com/file/d/1Gah652AQjtQP09CB0M2JIm2f_eIvZe1m/view?usp=sharing]
- **GitHub Repository Link:** [https://github.com/mufaddalbag5352-arch/EV-Market-Analysis]

1. Executive Summary

Transportation contributes approximately 24% of global CO_2 emissions, making electric vehicle (EV) adoption critical to global decarbonization. This project analyzes a comprehensive dataset of 30 electric vehicle models across 10 leading manufacturers, 6 manufacturing nations, 6 vehicle categories, and 5 battery technologies (LFP, NMC, NCA, Sodium-Ion, and Solid-State), representing **3,000,000 annual unit sales**. Using an interactive 9-visualization BI Dashboard and a 4-point Story, the solution addresses the analytical requirements of three core stakeholders.

2. Scenario-Wise Findings & Quantitative Insights

- **Scenario 1 — Emily Johnson (EV Market Strategy Director):**
  - **Category Contribution: SUVs** are the largest volume driver with **1,255,000 annual sales (41.83%)**, followed by **Sedans (725,000 units / 24.17%)** and **Hatchbacks (520,000 units / 17.33%)**.
  - **Manufacturer Ranking: BYD (785,000 units)** and **Tesla (640,000 units)** lead global volume, followed by **Volkswagen (340,000)**, **Hyundai (290,000)**, and **Ola Electric (265,000)**.
  - **Range vs. Satisfaction: Sedans** achieve the highest average driving range (**614.7 km**) and customer satisfaction (**4.70 / 5.0**), followed closely by **SUVs** (**492.3 km**, **4.63 / 5.0**).
- **Scenario 2 — David Chen (Automotive Business Intelligence Manager):**
  - **Market Share Concentration: BYD (26.17%)** and **Tesla (21.33%)** together account for **47.50%** of total global sales.
  - **Battery Technology Dominance: Lithium Iron Phosphate (LFP)** leads the market at **39.17% (1,175,000 units)** due to affordability and thermal stability, followed by **NMC at 31.33% (940,000 units)** and **NCA at 16.00% (480,000 units)**.

    - **Geographical Production: China** manufactures **29.50% (885,000 units)** of total volume, followed by the **United States (24.67% / 740,000 units)**, **Germany (16.83% / 505,000 units)**, and **India (15.50% / 465,000 units)**.
- **Scenario 3 — Dr. Sarah Williams (Sustainable Mobility Research Analyst):**
    - **Battery Efficiency Comparison: Solid-State** batteries deliver the highest average driving range (**745.0 km** from an average capacity of **120.3 kWh**), outperforming **NCA** (**593.4 km** at **193.6 kWh**) and **NMC** (**410.7 km** at **140.6 kWh**).
    - **Customer Satisfaction Heatmap:** Vehicles powered by **Solid-State** batteries record the highest average customer rating (**4.83 / 5.0**, peaking at **4.90** in Sedans), confirming that mitigating range anxiety directly drives consumer satisfaction.